

Lecture 2: Floating-Point Numbers

Rounding Error

1 Rounding Error

When we store a number on a computer we usually *do not keep its full value*. We keep only a finite floating-point approximation. The difference between the true value and the stored value is the **rounding error**.

Example (π). The true value is

$$\pi = 3.14159\underline{26535}\dots$$

If we keep only

$$\pi \approx 3.14,$$

then we have thrown away the underlined digits, so

$$|\pi - 3.14| > 0.$$

Example (Taylor series of $\sin$).

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \dots$$

For $x = 0.5$ the exact value is

$$\sin(0.5) = 0.479425\dots$$

- Taking only the first terms $x - \frac{x^3}{3!}$ gives about 0.47. Stopping early (not using the full/actual value) produces **rounding error**.
- The later terms $\left(-\frac{x^7}{7!} + \frac{x^9}{9!} - \dots\right)$ are the part we dropped: keeping fewer of them means *less precision / less correct*.

Rounding error as a number:

$$\underbrace{3.14159}_{\text{actual}} - \underbrace{3.14}_{\text{calculated value}} = \boxed{0.00159} \quad (\text{rounding error}).$$

Keeping only m digits

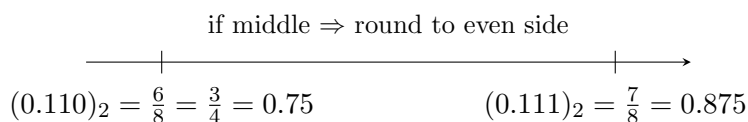
Suppose we keep only $m = 3$ digits after the point:

$$(0.d_1d_2d_3)_2 \quad (\text{only three places}).$$

Then a longer number must be rounded to that length, e.g.

$$(0.110001)_2 \approx (0.110)_2.$$

Tie-breaking rule: if the value falls exactly in the middle, round to the *even* side. For example $(0.1101)_2$ sits between $(0.110)_2$ and $(0.111)_2$, and a longer number such as $(0.1101101)_2$ must be rounded to one of the two endpoints.



2 Determining (Measuring) the Rounding Error

We want to know how “good or bad” the error is. A natural way is to compare the error to the size of the number itself:

$$\frac{\left| \overbrace{fl(x)}^{\text{rounded value}} - \overbrace{x}^{\text{actual value}} \right|}{|x|}.$$

Scale-invariant (relative) rounding error:

$$\delta = \frac{|fl(x) - x|}{|x|}.$$

This measures the *impact* of the rounding error, independent of scale.

Maximum scale-invariant rounding error / Machine Epsilon ε :

$\varepsilon =$ maximum scale-invariant rounding error.

This is the *tolerance* — the largest relative error we may suffer. We can never be 100% perfect, so ε tells us the worst case.

Note the proportionalities:

$$\delta_{\max} \propto \frac{1}{|x|}, \quad \delta_{\max} \propto |fl(x) - x|.$$

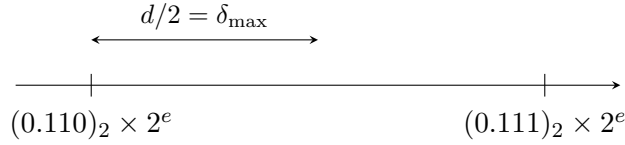
3 Standard Form and Machine Epsilon

Standard form (mantissa $0.d_1d_2d_3$)

$$\boxed{(0.d_1d_2d_3)_B \times B^e} \quad (m = 3)$$

Gap between two consecutive numbers (taking $B = 2$):

$$d = (0.001)_2 \times 2^e = (0 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3}) \times 2^e = 2^{-m} \times 2^e \approx B^{-m} B^e.$$



Maximum absolute error is half a gap:

$$\frac{d}{2} = \delta_m = |fl(x) - x|_{\max} = \frac{1}{2} B^{-m} B^e.$$

Smallest (normalized) $|x|$:

$$|x|_{\min} = (0.100)_2 \times 2^e = (1 \cdot 2^{-1}) \times 2^e = B^{-1} B^e.$$

Machine epsilon:

$$\varepsilon = \delta_{\max} = \frac{\frac{1}{2} B^{-m} B^e}{B^{-1} B^e} = \boxed{\frac{1}{2} B^{1-m}}.$$

The exponent e *cancels*, so it has no impact on ε / $\delta_{\max}$ / the machine epsilon.

De-normalized form (mantissa $1.d_1d_2d_3$)

$$\begin{aligned} & (1.d_1d_2d_3)_B \times B^e \\ |fl(x) - x|_{\max} &= \frac{1}{2} B^{-m} B^e, \quad |x|_{\min} = (1.000)_2 \times 2^e = (1)_2 \times 2^e = B^e, \\ \varepsilon &= \frac{\frac{1}{2} B^{-m} B^e}{B^e} = \frac{1}{2} B^{-m}. \end{aligned}$$

Normalized form (leading 1 bit fixed: $0.1 d_1d_2d_3$)

$$\begin{aligned} & \underbrace{(0.1 d_1d_2d_3)_B}_{\text{leading 1 bit}} \times B^e \\ |fl(x) - x|_{\max} &= \frac{1}{2} B^{-(m+1)} B^e, \quad |x|_{\min} = (0.1)_2 \times 2^e = B^{-1} B^e, \\ \varepsilon = \delta_{\max} &= \frac{|fl(x) - x|_{\max}}{|x|} = \frac{\frac{1}{2} B^{-(m+1)} B^e}{B^{-1} B^e} = \frac{1}{2} B^{-m}. \end{aligned}$$

4 How Rounding Error Occurs (Worked Example)

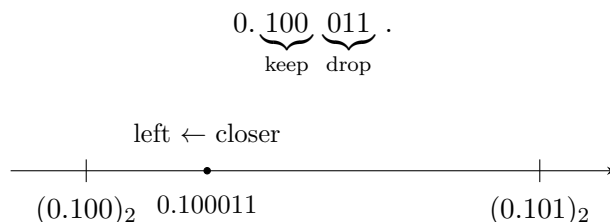
Take $B = 2$, $m = 3$, and

$$x = \frac{5}{8} = (0.101)_2, \quad y = \frac{7}{8} = (0.111)_2 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8}.$$

We want the **calculated** (true) product and the **stored** value $fl(x \cdot y)$:

$$x \cdot y = \frac{5}{8} \times \frac{7}{8} = \frac{35}{64} = \frac{32}{64} + \frac{2}{64} + \frac{1}{64} = \frac{1}{2} + \frac{1}{32} + \frac{1}{64} = 2^{-1} + 2^{-5} + 2^{-6} = (0.100011)_2.$$

This has more than $m = 3$ bits, so we must round:



The dropped part 011 is less than half a step, so we round *down*:

$$fl(x \cdot y) \approx (0.100)_2 = \frac{1}{2}.$$

The true product $\frac{35}{64}$ is replaced by $\frac{1}{2}$ — that difference is the rounding error.

5 Loss of Significance

When two nearly equal numbers are subtracted, the leading (significant) digits cancel and we are left with far fewer correct significant figures. Aligning

$$0.110 \quad \text{with} \quad 0.00100011 \times 2^{-2}$$

shows that the meaningful leading bits get lost in the subtraction.

Quadratic example

Solve

$$x^2 - 56x + 1 = 0, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

With $\sqrt{783} = 27.98$ (to 4 significant figures):

Exact-style (4 sig. figs.):

$$x_1 = 28 + \sqrt{783} = 55.98, \quad x_2 = 28 - \sqrt{783} = 0.01786.$$

On the computer (using $\sqrt{783} \approx 27.98$):

$$\begin{aligned} x_1 &= 28 + 27.98 = 55.98 \quad (\text{fine}), \\ x_2 &= 28 - 27.98 = 0.02000 \quad (\text{loss of significance}). \end{aligned}$$

The large root keeps its accuracy, but the small root loses almost all of its significant figures because of the cancellation.

A better way (avoiding cancellation)

Use the factored form

$$x^2 - (\alpha + \beta)x + \alpha\beta, \quad \alpha = x_1, \quad \beta = x_2.$$

Since the product of the roots is $\alpha\beta = \frac{c}{a} = 1$, we have

$$\beta = \frac{1}{\alpha} = \frac{1}{x_1} = x_2.$$

So compute the large root x_1 accurately first, then obtain the small root as $x_2 = 1/x_1$ instead of subtracting two nearly equal numbers.

6 Example: Rounding Before Computing

The true midpoint of 7.11 and 7.12 is

$$\frac{7.11 + 7.12}{2} = 7.115.$$

But if intermediate results are stored with only 3 significant figures:

$$fl\left(\frac{7.11 + 7.12}{2}\right) = fl\left(\frac{14.23}{2}\right) = fl\left(\frac{14.2}{2}\right) = 7.1.$$

Here 14.23 is first rounded to 14.2, so the final answer 7.1 differs from the true value 7.115. Rounding at an intermediate step changes the result.